Lab 7_SE

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1.Initial Run of Buggy **movie_tickets.py**

```
(.venv) PS C:\Users\amrit\OneDrive\Desktop\College\SE\SE_L6_L7_15_movie_tickets\Lab 7> Get-Content movie_tickets.py, README.md, test_movie_tickets.py
>> python movie_tickets.py
def parse_seat(seat):
    return seat[0].upper(), int(seat[1:])

def total_price(prices_csv):
    prices = [int(p) for p in prices_csv.split(',')]
    return round(sum(prices) * 1.18, 2)

def average_rating(ratings):
    values = [int(r) for r in ratings.split()]
    return round(sum(values) / len(values), 1)

def main():
    print("==== Running parse_seat ====")
    user_input = input("Enter a seat number (e.g. H12): ")
    result = parse_seat(user_input)
    print("Row and Seat:", result)

    print("\n==== Running total_price ====")
    user_input = input("Enter ticket prices as CSV (e.g. 150,250,400): ")
    result = total_price(user_input)
    print("Total with GST:", result)

    print("\n==== Running average_rating ====")
    user_input = input("Enter ratings separated by spaces (e.g. 4 5 3): ")
    result = average_rating(user_input)
    print("Average Rating:", result)

if __name__ == "__main__":
    main()
# L6_L7_15_movie_tickets â€" Movie Tickets

Lab 7: Fuzz Testing

## Files
```

```
Lab 7: Fuzz Testing

## Files

| File | Description |
|------|-------------|
| `movie_tickets.py` | Buggy code to test and fix |
| `test_movie_tickets.py` | Fuzz test file to complete using Hypothesis |

Refer to the lab handout for instructions.
from hypothesis import given, strategies as st
from movie_tickets import parse_seat, total_price, average_rating

@given(st.text() | st.none())
def test_parse_seat_no_crash(s):

@given(st.text() | st.none())
def test_total_price_no_crash(s):

@given(st.text() | st.none())
def test_average_rating_no_crash(s):
==== Running parse_seat ====
Enter a seat number (e.g. H12): H12
Row and Seat: ('H', 12)

==== Running total_price ====
Enter ticket prices as CSV (e.g. 150,250,400): 150,250,400
Total with GST: 944.0

==== Running average_rating ====
Enter ratings separated by spaces (e.g. 4 5 3): 4 5 3
Average Rating: 4.0
```

2. Failing Hypothesis Fuzz Tests (Bug Discovery)

```
========================= short test summary info =========================
FAILED test_movie_tickets.py::test_parse_seat_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
FAILED test_movie_tickets.py::test_total_price_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_movie_tickets.py::test_average_rating_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
========================== 3 failed in 8.03s ==========================
```

3. Run of Hardened **movie_tickets.py**

```
(.venv) PS C:\Users\amrit\OneDrive\Desktop\College\SE\SE_L6_L7_15_movie_tic
kets\Lab 7> python movie_tickets.py
==== Running parse_seat ====
Enter a seat number (e.g. H12): H12
Row and Seat: ('H', 12)

==== Running total_price ====
Enter ticket prices as CSV (e.g. 150,250,400): 150,250,400
Total with GST: 944.0

==== Running average_rating ====
Enter ratings separated by spaces (e.g. 4 5 3): 4 5 3
Average Rating: 4.0
```

```
(.venv) PS C:\Users\amrit\OneDrive\Desktop\College\SE\SE_L6_L7_15_movie_tic
kets\Lab 7> python movie_tickets.py
==== Running parse_seat ====
Enter a seat number (e.g. H12): A23
Row and Seat: ('A', 23)

==== Running total_price ====
Enter ticket prices as CSV (e.g. 150,250,400): 23456A
Total with GST: 0.0

==== Running average_rating ====
Enter ratings separated by spaces (e.g. 4 5 3): 10
Average Rating: 10.0
```

4. Passing Hypothesis Fuzz Tests (Verification)

```
(.venv) PS C:\Users\amrit\OneDrive\Desktop\College\SE\SE_L6_L7_15_movie_tic
kets\Lab 7> python -m pytest test_movie_tickets.py
========================== test session starts ==========================
platform win32 -- Python 3.14.0, pytest-8.0.0, pluggy-1.7.0
rootdir: C:\Users\amrit\OneDrive\Desktop\College\SE\SE_L6_L7_15_movie_ticke
ts\Lab 7
plugins: hypothesis-6.170.15
collected 3 items

test_movie_tickets.py ...                                          [100%]

========================== 3 passed in 1.63s ===========================
```